Mano Engineering LLC
2512 Northland Drive, Mendota Heights, MN 55120

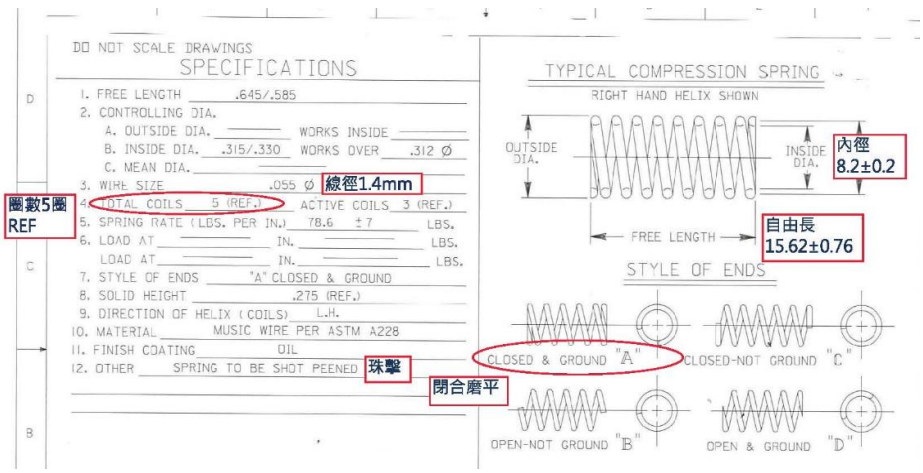
出貨檢查記錄表 Inspection Report

產品編號 MK's P/N	Z0064-003	料號 Customer P/N	F004
生產批號 Lot No.	P-02601051	品名 Part Name	SPRING, COMPRESSION

檢驗日期 Date	February 23, 2026			
批量 Lot Size	500			
抽樣標準 Sampling Criteria	允收 Accept	0	拒收 Reject	1
抽樣數 Sample Size	5			

檢驗量具代號 INSPECTION GAUGES

1.游標卡尺Vernier Caliper 2.分厘卡Micrometer 3.捲尺Measuring Tape
4.拉壓力測試機Spring Force Tester 5.扭力荷重機Torque Tester 6.塞規/圓棒Pin Gauge
7.治具Fixture 8.高度規Height Gauge 9.硬度計Hardness Tester 10. 投影機Projector
11.目視Visual 12.其他Others



項目 Item	檢驗項目 Description	規格 Spec. & Tol.	量具 Gauge Code	檢驗結果 Data 1	2	3	4	5	6	7	8	9	10
1	Material	SWPB	12	OK	OK	OK	OK	OK					
2	Wire Dia.	1.4	2	1.4	1.4	1.4	1.4	1.4					
3	Inside Dia.	8.2±0.2	1	8.25	8.28	8.29	8.25	8.26					
4	Free Length	15.62±0.76	1	15.62	15.47	15.49	16.55	16.58					
5	Total Coils	(5T)	11	5	5	5	5	5					
6	Wind	Left Hand	11	L	L	L	L	L					
7	Finish	Oil	11	OK	OK	OK	OK	OK					
8	Ground	Ends Closed and Ground	11	OK	OK	OK	OK	OK					
9	Rate	78.6±7.0LBS (1403±125gf/mm)	4										
10													
11													
12													
13													

Unit: millimeter (mm)

No. 1
1 14.00 mm 1906.0 gf
2 12.50 mm 3908.0 gf
K 1334.67 gf/mm
No. 2
1 14.00 mm 2011.0 gf
2 12.50 mm 4013.0 gf
K 1334.67 gf/mm
No. 3
1 14.00 mm 2013.0 gf
2 12.50 mm 4033.0 gf
K 1346.67 gf/mm
No. 4
1 14.00 mm 1954.0 gf
2 12.50 mm 3970.0 gf
K 1344.00 gf/mm
No. 5
1 14.00 mm 1911.0 gf
2 12.50 mm 3918.0 gf
K 1338.00 gf/mm

主管 Supervisor	檢驗 Inspector	判定Judgement	處理意見Comments:
品管部 2026.02.23 黃嫻穎	品管部 2026.02.23 陳怡伶	合格 OK	
		不合格 NG	

Mano Engineering LLC

Address: 2512 Northland Drive, Mendota Heights, MN 55120

MILL TEST CERTIFICATE / CERTIFICATE OF INSPECTION

Certificate No : 202508-1268P2-3 (1 / 4)
CA No/Coil No : 1-29

Issue Date : Aug. 25. 2025
P.O No : -
L/C No. : 5AQQZ2000034-523 AUG.22,25
Invoice No : VH1268P2

1. TYPE & GRADE

Diameter: 1.400 mm
Usage: SPRING
Total Net Weight: 2039.00 kgs

Coating: UNGALVANIZED
Grade: SWP-B
Lubrication: PHOSPHATED, LIGHT OILED
Specification: JIS G3522

2. TEST RESULTS

Item		Diameter ±(mm)	Tensile Strength (N/mm2)	Torsion (Turns)	Area Reduction (Min %)	Wrapping / Coiling	Bending	Weight of Coating(g/m2)
Spec		0.015	2160 - 2350	25	-	-	-	-
Test No	1	1.393	2,226	30	-	-	-	-
	2	1.393	2,226	30	-	-	-	-
	3	1.393	2,226	30	-	-	-	-
	4	1.393	2,226	30	-	-	-	-
	5	1.393	2,226	30	-	-	-	-
	6	1.393	2,239	35	-	-	-	-
	7	1.393	2,239	35	-	-	-	-
	8	1.393	2,239	35	-	-	-	-
	9	1.393	2,239	35	-	-	-	-

We hereby certify that the above results are true and correct in every detail.

CHEMICAL ANALYSIS OF WIRE ROD

Lead and Mercury Free

RoHS Compliant

Charge No.	Chemical Composition (%)					
	C	Si	Mn	P	S	Cu
Specification	0.80~0.85	0.12~0.32	0.30~0.60	MAX 0.025	MAX 0.025	MAX 0.20
SF 89500	0.82	0.18	0.39	0.010	0.008	0.014

Approval or Inspection Possibility

WIRE ROPE
Suncheon Factory : 15, Sandan 1-gil, Seo-myeon, Suncheon-si Jeonranam-do, Korea
Yulchon Factory : 80-21, Yulchonsandan 1-ro, Haeryong-myeon, Suncheon-si, Jeonranam-do, Korea

WIRE
Suncheon 2nd Factory : 38, Sandan 4-gil, Seo-myeon, Suncheon-si Jeonranam-do, Korea
Yulchon Factory : 80-21, Yulchonsandan 1-ro, Haeryong-myeon, Suncheon-si, Jeonranam-do, Korea

ABS LLOYD'S NK KR DNV·GL BV RINA CCS RS

Quality Management Dept.

Chief Inspector

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Spec		0.015	2160 - 2350	25	-	-	-	-
Test No	10	1.393	2,239	35	-	-	-	-
	11	1.394	2,229	34	-	-	-	-
	12	1.394	2,229	34	-	-	-	-
	13	1.394	2,229	34	-	-	-	-
	14	1.394	2,229	34	-	-	-	-
	15	1.394	2,229	34	-	-	-	-
	16	1.394	2,227	36	-	-	-	-
	17	1.394	2,227	36	-	-	-	-
	18	1.394	2,227	36	-	-	-	-

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2. TEST RESULTS

Item		Diameter ±(mm)	Tensile Strength (N/mm2)	Torsion (Turns)	Area Reduction (Min %)	Wrapping / Coiling	Bending	Weight of Coating(g/m2)
Spec		0.015	2160 - 2350	25	-	-	-	-
Test No	19	1.394	2,227	36	-	-	-	-
	20	1.394	2,227	36	-	-	-	-
V	21	1.393	2,231	31	-	-	-	-
V	22	1.393	2,231	31	-	-	-	-
	23	1.393	2,231	31	-	-	-	-
	24	1.393	2,231	31	-	-	-	-
	25	1.393	2,231	31	-	-	-	-
	26	1.394	2,220	32	-	-	-	-
	27	1.394	2,220	32	-	-	-	-

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Spec		0.015	2160 - 2350	25	-	-	-	-
Test No	28	1.394	2,220	32	-	-	-	-
	29	1.394	2,220	32	-	-	-	-
	30							
	31							
	32							
	33							
	34							
	35							
	36							

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